Database Compression Using Information Theory

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Abstract

This skeleton demonstrates how to use the `infthesis` style for undergraduate dissertations in the School of Informatics. It also emphasises the page limit, and that you must not deviate from the required style. The file `skeleton.tex` generates this document and should be used as a starting point for your thesis. Replace this abstract text with a concise summary of your report.

Research Ethics Approval

This project was planned in accordance with the Informatics Research Ethics policy. It did not involve any aspects that required approval from the Informatics Research Ethics committee.

Declaration

I declare that this thesis was composed by myself, that the work contained herein is my own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified.

(*Finlay Peden*)

Acknowledgements

Any acknowledgements go here.

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Chapter 1

Introduction

The preliminary material of your report should contain:

- The title page.
- An abstract page.
- Declaration of ethics and own work.
- Optionally an acknowledgements page.
- The table of contents.

As in this example `skeleton.tex`, the above material should be included between:

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    ...
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```

This style file uses roman numeral page numbers for the preliminary material.

The main content of the dissertation, starting with the first chapter, starts with page 1. ***The main content must not go beyond page 40.***

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1.1 Using Sections

Divide your chapters into sub-parts as appropriate.

1.2 Citations

When citing work using author names, your sentences should still read as correct English if any parts in parenthesis are removed. Use `\citet` when citing the name as text that's part of your sentence:

> We follow the method of Arimura [1].

and use `\citep` for a parenthetical citation:

> It's possible to learn first-order Horn sequences from entailment [4].

You may use any consistent reference style that you prefer, including "[1]" numerical citations.

Chapter 2

Background

In this chapter, we will give a general knowledge of the methods needed to understand our proposed compression approach and how to measure its effectiveness. We will start by discussing acyclic schemas along with Multivalued Dependencies (MVDs) and how they will be used to compress the data. We will then explain how the definition of entropy from information theory can be used in determining data loss during the compression. Then we summarize the work by Kenig et al. [5] in which they describe systems to discover MVDs and Acyclic Join Dependencies. Finally, we will briefly describe widely used database compression techniques.

To keep consistency we will use the same notation as Kenig et al. [5]. Let Ω be a set of variables, also called attributes. If $X, Y \subseteq \Omega$, then XY denotes $X \cup Y$.

2.1 Running Example

We will use (Figure 2.1) as an example to demonstrate the information about the background topics. We consider the relation R over the schema $\Omega = \{A, B, C, D\}$. As illustrated, the table R can be decomposed into smaller tables, and joining these tables reconstructs R. The join dependency is $R = R[AB] \bowtie R[AC] \bowtie R[BD]$. This dependency only holds for the rows in black. The red rows will be used as an example of approximate acyclic schemas in Section 2.4.

R:

A	B	C	D	
a_1	b_1	c_1	d_1	1/4
a_1	b_2	c_1	d_2	1/4
a_2	b_1	c_2	d_1	1/4
a_2	b_2	c_2	d_2	1/4
a_1	b_2	c_2	d_2	

$=$

A	B	
a_1	b_1	1/4
a_1	b_2	1/4
a_2	b_1	1/4
a_2	b_2	1/4

$\bowtie$

A	C	
a_1	c_1	1/2
a_2	c_2	1/2
a_1	c_2	

$\bowtie$

B	D	
b_1	d_1	1/2
b_2	d_2	1/2

Figure 2.1: A relation R and its decomposition into an acyclic schema

2.2 Acyclic Schemas and Multivalued Dependencies

In this section, ignore the probabilities (discussed in Sec. 2.3) and the red rows (discussed in Sec. 2.4) in Figure 2.1.

For $Y \subseteq \Omega$ we let $R[Y]$ denote the projection of R onto the attributes Y.

When trying to compress databases, it is often useful to decompose a table R with attributes Ω into smaller tables. A schema is a set $S = \{\Omega_1, \ldots, \Omega_k\}$ of attribute sets whose union is Ω, where no set is contained in another, since a contained set adds no information to the join. R satisfies the join dependency $\mathrm{JD}(S)$, written $R \models \mathrm{JD}(S)$, when joining the projections $R[\Omega_1], \ldots, R[\Omega_k]$ gives us exactly the table R. This also means that the decomposition is lossless as joining the projections gives exactly R.

An MVD is a special type of join dependency in which, for each value of a set of attributes X, the remaining attributes split into groups that vary independently of one another. Let $X, Y_1, \ldots, Y_m \subseteq \Omega$. We say R satisfies the multivalued dependency $X \twoheadrightarrow Y_1|Y_2|\ldots|Y_m$ if $R = R[XY_1] \bowtie \cdots \bowtie R[XY_m]$. Where $m \geq 2$, the Y_is are pairwise disjoint and $XY_1 \cdots Y_m = \Omega$. We call X the *key* and $\{Y_1, \ldots, Y_m\}$ the *dependents*. When $m = 2$ we call it a *standard MVD* which is the most common form in the literature.

A schema is *acyclic* if it has a join tree. The schema $\{AB, AC, BD\}$ of Figure 2.1 is acyclic, as shown by the join tree in Figure 2.2.

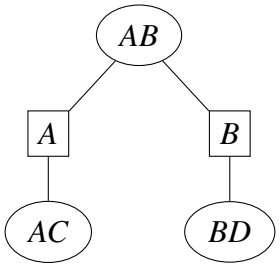


Figure 2.2: Join Tree

An acyclic schema can be fully described by a set of MVDs, one for each edge of the join tree. The acyclic schema can be described by these two multivalued dependencies: $B \twoheadrightarrow D|AC$, $A \twoheadrightarrow C|BD$. For example the edge $AB \overset{\mathrm{B}}{-} BD$ in the join tree defines the MVD $B \twoheadrightarrow D|AC$. The *separator* of an edge is the intersection of the two bags it joins. Removing an edge splits the tree in two, and the attributes outside the separator split accordingly. The separator and these two groups give an MVD whose key is the separator.

The *separator* of an edge of a join tree is the intersection of the two bags it joins. Removing the edge splits the tree into two subtrees, and the attributes of each subtree that are not in the separator form the two dependents of an MVD whose key is the separator. In Figure 2.2, the edge between AB and BD has separator B and gives the MVD $B \twoheadrightarrow D|AC$, and the edge between AB and AC has separator A and gives the MVD $A \twoheadrightarrow C|BD$. Beeri et al. [3] showed that R satisfies the acyclic join dependency of the schema exactly when it satisfies the MVD of every edge.

A *join tree* for a schema S is an undirected tree with a set of nodes $\Omega_1, \ldots, \Omega_k$ of S, where the following properties hold: every edge of the graph (Ω_i, Ω_j) must be labelled with the set of attributes $\Omega_i \cap \Omega_j$ and for all attributes A in $\Omega_i \cap \Omega_j$ each edge along a path between the pair of distinct nodes $\Omega_i, \Omega_j (\Omega_i \neq \Omega_j)$ must include the label A [3].

We say that R satisfies the *acyclic join dependency* AJD(S), written $R \models \text{AJD}(S)$, if the schema S is acyclic and $R \models \text{JD}(S)$. A single MVD $X \twoheadrightarrow Y_1 | \cdots | Y_m$ is the simplest case, since it is the join dependency of the schema $\{XY_1, \ldots, XY_m\}$, whose sets all intersect in X. This also explains why one MVD with several dependents is a compact way of writing several standard ones: for example, $X \twoheadrightarrow A|B|C$ holds exactly when $X \twoheadrightarrow AB|C$, $X \twoheadrightarrow A|BC$ and $X \twoheadrightarrow AC|B$ all hold [2].

The set of MVDs obtained from the edges of a join tree $\mathcal{T}$ is called the *support* of $\mathcal{T}$. Since a tree with k nodes has $k-1$ edges, the support contains $k-1$ MVDs. Because no set of S may be contained in another, an acyclic schema over n attributes has at most n sets [3], so a decomposition of R into an acyclic schema never produces more tables than R has attributes. In Figure 2.2 the support consists of two MVDs and the schema has three sets, which is less than the 4 attributes of R.

2.3 Information Theory

Information theory is the mathematics of quantifying and measuring information. In our context we will be using it to measure the uncertainty in data. Entropy is the main component of information theory that is relevant to this project. It is used to calculate how uncertain a random variable is.

Let X be a discrete random variable with a set of possible values $\mathcal{X}$ and probability mass function $p(x)$ such that the sum of all probabilities is equal to one. The entropy of X is defined as:

$$H(X) = \sum_{x \in \mathcal{X}} p(x) \log \frac{1}{p(x)} \tag{2.1}$$

Mutual Information is the measure of mutual dependence between two random variables. It equals 0 if an only if the two variables are independent. We write it as:

$$I(B;C|A) = H(AB) + H(AC) - H(ABC) - H(A) \tag{2.2}$$

Following the work done by Lee [6, 7] as presented by Kenig et al. [5]. Let $\mathcal{T}$ be a join tree. We define the following expression:

$$J(S) = \sum_{i=1}^{n} H(\Omega_i) - \sum_{(\Omega_i, \Omega_j) \in \text{edges}(\mathcal{T})} H(\Omega_i \cap \Omega_j) - H(\Omega) \tag{2.3}$$

2.4 Mining approximate acyclic schemes from relation

MAIMON [5](Multivalued Approximate Inference Mining and NOrmalization) is the system proposed by Kenig et al. to discover MVDs and acyclic schemes in data. They

describe enumeration algorithms that allow these discoveries. They also describe a method of calculating the joint entropy of a group of attributes. When evaluating their work they found that this method could be scaled to 1 million rows and 30 attributes with small amounts of spurious tuples. Where a spurious tuple is a row is a tuple not in the original database. Computing the entropy of a set of attributes is the most computationally expensive part of the MAIMON system. Their use of approximate MVDs allowed them to mark more schema as suitable for the compression leading to the compression working even when an exact MVD could not be found. If we now look at the red rows in Figure 2.1 we can see that by adding the row to the original table there now needs to be a new row in the BDE table. However this now means that when the tables are joined together there will be spurious table that was not in the original table.

If we now look at the red rows in the running example we can see that if we add the 5th red row to R then we also need to add the row a_1, c_1 in $R[AC]$ as well. However this leads to a spurious tuple in our final result after the joins being a new tuple (a_1, b_1, c_2, d_1). In the paper they observe this and note that if only perfect MVDs and acyclic schema were taken then it would not apply widely to databases as if one row in the database changed then the whole schema could no longer be valid. Therefore they chose to use approximate MVDs and acyclic schema instead.

2.5 Database Compression Techniques

In this section we will give a brief overview of some general database compression techniques that give a better understanding of the topic of database compression and give an option to compress our data further when it can no longer be decomposed using the techniques in [5].

Chapter 3

Conclusions

3.1 Final Reminder

The body of your dissertation, before the references and any appendices, *must* finish by page 40. The introduction, after preliminary material, should have started on page 1.

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Bibliography

[1] Hiroki Arimura. Learning acyclic first-order horn sentences from entailment. In *Proc. of the 8th Intl. Conf. on Algorithmic Learning Theory, ALT '97*, pages 432–445, 1997.

[2] Catriel Beeri, Ronald Fagin, and John H. Howard. A complete axiomatization for functional and multivalued dependencies in database relations. In Diane C. P. Smith, editor, *Proceedings of the 1977 ACM SIGMOD International Conference on Management of Data, Toronto, Canada, August 3-5, 1977*, pages 47–61. ACM, 1977. doi: 10.1145/509404.509414. URL https://doi.org/10.1145/509404.509414.

[3] Catriel Beeri, Ronald Fagin, David Maier, and Mihalis Yannakakis. On the desirability of acyclic database schemes. *J. ACM*, 30(3):479–513, 1983. doi: 10.1145/2402.322389. URL https://doi.org/10.1145/2402.322389.

[4] Chen-Chung Chang and H. Jerome Keisler. *Model Theory*. North-Holland, third edition, 1990.

[5] Batya Kenig, Pranay Mundra, Guna Prasaad, Babak Salimi, and Dan Suciu. Mining approximate acyclic schemes from relations. In David Maier, Rachel Pottinger, AnHai Doan, Wang-Chiew Tan, Abdussalam Alawini, and Hung Q. Ngo, editors, *Proceedings of the 2020 International Conference on Management of Data, SIGMOD Conference 2020, online conference [Portland, OR, USA], June 14-19, 2020*, pages 297–312. ACM, 2020. doi: 10.1145/3318464.3380573. URL https://doi.org/10.1145/3318464.3380573.

[6] Tony T. Lee. An information-theoretic analysis of relational databases - part I: data dependencies and information metric. *IEEE Trans. Software Eng.*, 13(10):1049–1061, 1987. doi: 10.1109/TSE.1987.232847. URL https://doi.org/10.1109/TSE.1987.232847.

[7] Tony T. Lee. An information-theoretic analysis of relational databases - part II: information structures of database schemas. *IEEE Trans. Software Eng.*, 13(10):1061–1072, 1987. doi: 10.1109/TSE.1987.232848. URL https://doi.org/10.1109/TSE.1987.232848.

Appendix A

First appendix

A.1 First section

Any appendices, including any required ethics information, should be included after the references.

Markers do not have to consider appendices. Make sure that your contributions are made clear in the main body of the dissertation (within the page limit).

Appendix B

Participants' information sheet

If you had human participants, include key information that they were given in an appendix, and point to it from the ethics declaration.

Appendix C

Participants' consent form

If you had human participants, include information about how consent was gathered in an appendix, and point to it from the ethics declaration. This information is often a copy of a consent form.